The documentation process conversion measures necessary to comply with this revision shall be completed by 28 August 1997

INCH POUND

MIL-PRF-19500/611A
28 May 1997
SUPERSEDING
MIL-S-19500/611
30 July 1993

PERFORMANCE SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER SWITCHING
TYPES 2N7374, 2N7375, 2N7376, AND 2N7377, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, power transistors for use in high-speed power switching applications. Three levels of product assurance are provided as specified in MIL-PRF-19500.

1.2 Physical dimensions. For 2N7376 and 2N7377, see figure 1 (TO-254AA), for 2N7374 and 2N7375, see figure 2 (TO-257AA).

1.3 Maximum ratings.

Types	P_T T_A = +25°C	P_T T_C = +100°C	V_{EBO}	V_{CBO}	V_{CEO}	I_C	I_B	T_J and T_{STG}	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W Max
2N7374, 2N7376	2.5 1/	30 2/	6	250	200	5.0	1	-65 to +200	3.3
2N7375, 2N7377	2.5 1/	30 2/	6	400	300	5.0	1	-65 to +200	3.3

1/ Derate linearly 14.3 mW/°C above $T_A \geq$ +25°C.
2/ Derate linearly 300 mW/°C above $T_C \geq$ +100°C.

1.4 Primary electrical characteristics.

	h_{FE}		h_{fe}	$V_{BE(SAT)}$ 1/	$V_{CE(SAT)}$ 1/	Pulse response		
	V_{CE} = 5.0 V dc I_C = 1.0 A dc		V_{CE} = 5.0 V dc I_C = 0.5 A dc f = 10 MHz	I_C = 3.0 A dc	I_C = 3.0 A dc	t_{on} I_C = 1 A dc	t_{off} I_C = 1 A dc	
	2N7374 2N7376	2N7375 2N7377					2N7374 2N7376	2N7375 2N7377
				V dc	V dc	µs	µs	µs
Min	40	25	2.0					
Max	120	75	7.0	1.2	0.4	0.25	1.5	2.0

1/ I_B = 0.3 A dc for 2N7374 and 2N7376; I_B = 0.6 A dc for 2N7375 and 2N7377.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified on figure 1 herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages. The US government's preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.3.1 Lead formation and finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition requirements (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14, of MIL-PRF-19500.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Screening (see 4.3)
c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CES1} and h_{FE2}	Not applicable
11	Subgroup 2 of table I herein; I_{CES1} and h_{FE2}; ΔI_{CES1} = 100% of initial value or 20 nA dc whichever is greater. $\Delta h_{FE2} = \pm 15\%$ of initial value.	I_{CES1} and h_{FE2}
12	t = 160 hours	t = 80 hours
13	Subgroups 2 and 3 of table I herein; I_{CES1} and h_{FE2}; ΔI_{CES1} = 100% of initial value or 20 nA dc whichever is greater. $\Delta h_{FE2} = \pm 15\%$ of initial value.	Subgroup 2 of table I herein; I_{CES1} and h_{FE2}; ΔI_{CES1} = 100% of initial value or 20 nA dc whichever is greater. $\Delta h_{FE2} = \pm 15\%$ of initial value.

1/ May be performed anytime before screen 9.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

$T_J = 187.5 \pm 12.5°C$, $V_{CE} \geq 20$ V dc, $T_A \leq 100°C$

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3131. The maximum limit (not to exceed the Group A, Subgroup 2 limit) for $Z_{\theta JX}$ in screening (appendix E, table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.3.2.1 For initial qualification or requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum). Twenty-two serialized devices shall be sent to the qualifying activity for test correlation.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, end point measurements: $Z_{\theta JX}$ = 3.1°C/W.

a.	I_M	10 mA.
b.	V_{CE} measurement voltage	20 V (same as V_H).
c.	I_H collector heating current	1 A (minimum).
d.	V_H collector-emitter heating voltage	20 V (minimum).
e.	t_H heating time	100 ms.
f.	t_{MD} measurement delay time	50 µs to 80 µs.
g.	t_{SW} sample window time	10 µs (maximum).

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, tables VIa and VIb of MIL-PRF-19500. Electrical measurements (end points) shall be in accordance with the applicable steps and table I, group A, subgroup 2 herein except $Z_{\theta JX}$, shall be performed after Intermittent Life (subgroup 4) only.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 30 V dc; P_T = 2.5 W at T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling on the devices shall be permitted.
B5	1027	See 4.5.4.
B6	3131	See 4.5.2

4.4.2.2 Group B inspection, appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1037	V_{CB} = 30 V dc; P_T = 2.5 W at T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced aircooling on the devices shall be permitted.
B6	3131	See 4.5.2

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end points) shall be in accordance with the applicable steps and table I, group A, subgroup 2 herein except $Z_{\theta JX}$, shall be performed after Intermittent Life (subgroup 4) only.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition A; weight 3 pounds; time 15 seconds.
C6	1037	V_{CB} = 30 V dc minimum; P_T = 2.5 W at T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750; $t_{on} = t_{off}$ = 3 minutes minimum for 6,000 cycles. No heat sink or forced air cooling on the devices shall be permitted.

4.4.4 Group E inspection. Group e inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 Electrical measurements (end points) shall be in accordance with the applicable steps and table I, group A, subgroup 2 herein except $Z_{\theta JX}$ shall not be performed.

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4.4.4 Group E inspection. Group e inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 Electrical measurements (end points) shall be in accordance with the applicable steps and table I, group A, subgroup 2 herein except $Z_{\theta JX}$ shall not be performed.

4.4.4.1 Group E inspection, appendix E, table IX of MIL-PRF-19500.

Subgroup	Method	Condition	Sampling Plan
E1	1051	500 cycles	22 devices, c = 0
E2	1039	Condition A, 500 hours	22 devices, c = 0
E3		Not applicable	
E4	3131	See 4.4.1	10 devices, c = 0
E5	1001	2N7374, 2N7376 - V_{CB} = 250V 2N7375, 2N7377 - V_{CB} = 460V P = 8 ± 2mm Hg for 60 s minimum	15 devices, c = 0

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 0.833 A dc for both device types.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be $25°C \leq T_R \leq 75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 3.3°C/W.

4.5.3 Inspection conditions. Unless otherwise specified in MIL-PRF-19500 or herein, all inspections shall be conducted at a case temperature (T_C) of +25°C ± 3°C.

4.5.4 Group B accelerated life test. This test shall be conducted using one of the two options listed herein (a and b) with the following conditions applying to all options: V_{CB} = 20 V minimum dc; 96 hours minimum;
T_J = +275°C.

a. P_T adjusted to give a lot average of T_J = +275°C with T_A = +125°C ± 25°C.

b. T_A = +25°C ± 3°C with P_T adjusted to give a lot average of T_J = +275°C.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		3.1	°C/W
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 10 mA dc; Pulsed (see. 4.5.1)	$V_{(BR)CEO}$			V dc
2N7374, 2N7376				200		
2N7375, 2N7377				300		
Collector to emitter breakdown voltage	3011	Bias condition B; R_1 = 100 Ω; I_C = 10 mA dc; Pulsed (see. 4.5.1)	$V_{(BR)CER}$			V dc
2N7374, 2N7376				250		
2N7375, 2N7377				400		
Emitter to base breakdown voltage	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	6		V dc
Collector to emitter cutoff current	3041	Bias condition C	I_{CES1}		0.2	µA dc
2N7374, 2N7376		V_{CE} = 200 V dc				
2N7375, 2N7377		V_{CE} = 300 V dc				
Collector to base cutoff current	3041	Bias condition D	I_{CBO}			
2N7374, 2N7376		V_{CB} = 200 V dc			0.1	µA dc
		V_{CB} = 250 V dc			1.0	µA dc
2N7375, 2N7377		V_{CB} = 300 V dc			0.1	µA dc
		V_{CB} = 400 V dc			1.0	µA dc
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 0.5 A dc; Pulsed (see 4.5.1)	h_{FE1}			
2N7374, 2N7376				40		
2N7375, 2N7377				25		

See footnote at end of table.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; Pulsed (see 4.5.1)	h_{FE2}			
2N7374, 2N7376				40	120	
2N7375, 2N7377				25	75	
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 3.0 A dc; Pulsed (see 4.5.1)	h_{FE3}			
2N7374, 2N7376				15		
2N7375, 2N7377				10		
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 5.0 A dc; Pulsed (see 4.5.1)	h_{FE4}	5		
Base to emitter saturated voltage	3066	Test condition A; I_C = 3.0 A dc	$V_{BE(sat)1}$		1.2	V dc
2N7374, 2N7376		I_B = 0.3 A dc; Pulsed (see 4.5.1)				
2N7375, 2N7377		I_B = 0.6 A dc; Pulsed (see 4.5.1)				
Collector to emitter saturated voltage	3071	I_C = 5 A dc; I_B = 1.0 A dc; Pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.0	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 5.0 A dc I_B = 1.0 A dc Pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.5	V dc
Collector-emitter voltage (saturated)	3071	I_C = 3.0 A dc	$V_{CE(sat)1}$		0.4	V dc
2N7374, 2N7376		I_B = 0.3 A dc; pulsed (see 4.5.1)				
2N7375, 2N7377		I_B = 0.6 A dc; pulsed (see 4.5.1)				
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C;	I_{CES2}		100	µA dc
2N7374, 2N7376		V_{CE} = 200 V dc				
2N7375, 2N7377		V_{CE} = 300 V dc				

See footnote at end of table.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; Pulsed (see 4.5.1)	h_{FE5}			
2N7374, 2N7376				12		
2N7375, 2N7377				8		
Subgroup 4						
Magnitude of small-signal short- circuit forward- current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 0.5 A dc; f = 10 MHz	$\|h_{fe}\|$	2.0	7.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		120	pf
Pulse response:						
Turn-on time	3251	Test condition A; I_C = 1.0 A dc; V_{CC} = 100 V dc	t_{on}		0.25	µs
2N7374, 2N7376		See figure 3				
2N7375, 2N7377		See figure 4				
Turn-off time	3251	Test condition A; I_C = 1.0 A dc; V_{CC} = 100 V dc	t_{off}			µs
2N7374, 2N7376		See figure 3			1.5	
2N7375, 2N7377		See figure 4			2.0	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +100°C, t ≥ 1 s; 1 cycle $t_r + t_f$ = 10 µs (See figure 5)				
Test 1 - All		V_{CE} = 6 V dc; I_C = 5 A dc				
Test 2 - All		V_{CE} = 40 V dc; I_C = 0.75 V dc				
Test 3 - 2N7374, 2N7376		V_{CE} = 200 V dc; I_C = 43 mA dc				
Test 4 - 2N7375, 2N7377		V_{CE} = 300 V dc; I_C = 21 mA dc				

See footnote at end of table.

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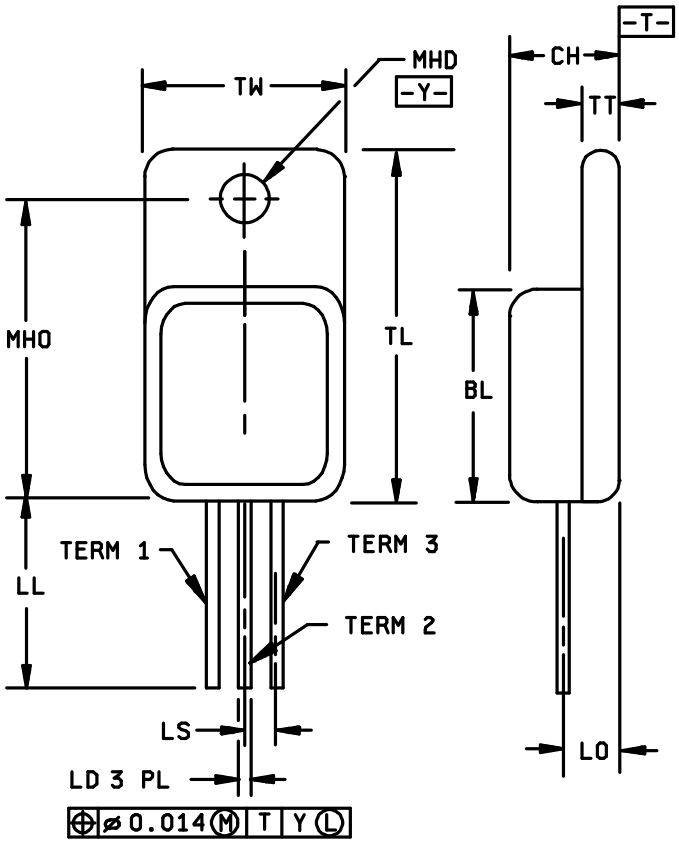
TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (switching)	3053	Load condition B (clamped inductive load) (see figure 6); T_C = +100°C; $t_r + t_f \leq 10$ µs; duty cycle ≤ 2%; t_p = 4 ms; R_S = 0.5 ohms; R_{BB1} = 50 ohms; V_{BB1} = 50 V R_{BB1} = 50 ohms; V_{BB1} = 50 V dc; R_{BB2} = 50 ohms; V_{BB2} = -4 V dc; I_C = 5 A dc; V_{CC} = 50 V dc; $R_L \leq 2.5$ Ω; L = 40 mH (Triad C-48U or equivalent).				
2N7374, 2N7376		Clamp voltage = 200 +0, -5 V dc				
2N7375, 2N7377		Clamp voltage = 300 +0, -5 V dc				
Safe operating area (unclamped inductive) Test 1	3053	Load condition C (unclamped inductive load) (see figure 7); T_C = +100°C; $t_r + t_f \leq 10$ µs; duty cycle ≤ 2%; t_p = 50 µs (vary obtain I_C); R_S = 0.12 ohms; R_{BB1} = 50 ohms; V_{BB1} = 50 V dc, R_{BB2} = 100 ohms; V_{BB2} = 0.5 V dc; I_C = 5 A dc; V_{CC} = 12.5 V dc; R = 0.33 Ω; L = .065 mH (20 turns 1 row of no. 20 AWG wire on Arnold Engineering Co. core no. A-930157-2, or equivalent)				
Electrical measurements		Group A, subgroup 2				

1/ For sampling plan, see MIL-PRF 19500.

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Ltr	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
CH	6.32	6.60	.249	.260
TT	1.02	1.27	.040	.050
LO	3.81 BSC		.150 BSC	
TL	20.07	20.32	.790	.800
BL	13.59	13.89	.535	.545
LS	3.81 BSC		.150 BSC	
TW	13.59	13.89	.535	.545
LL	30.35	31.37	1.195	1.235
MHO	16.89	17.40	.665	.685
MHD	3.53	3.78	.139	.149
LD	0.89	1.43	.035	.045
Term 1	Base			
Term 2	Collector			
Term 3	Emitter			

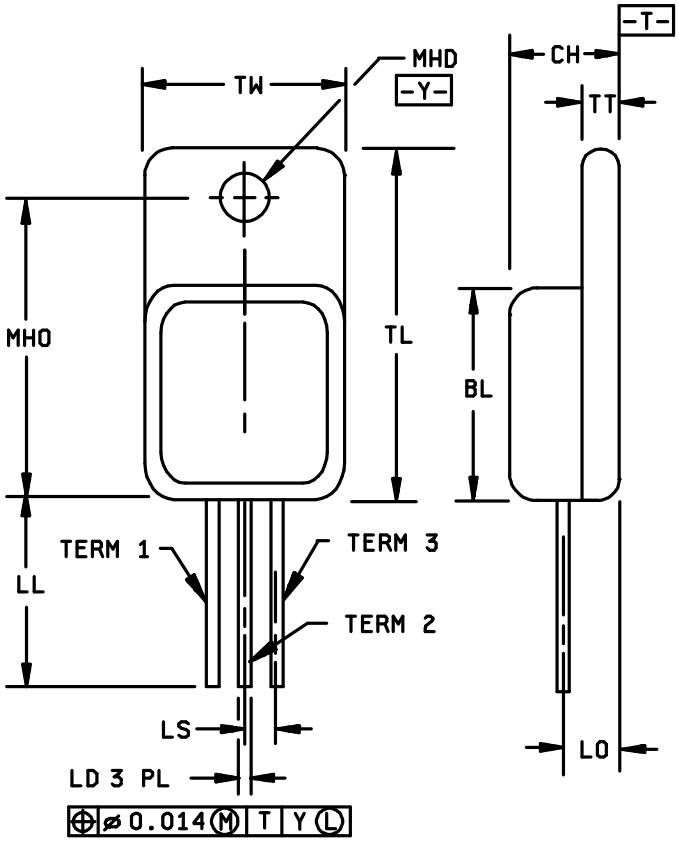


NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. All terminals are isolated from case.

FIGURE 1. Dimensions and configuration (T0-254AA).

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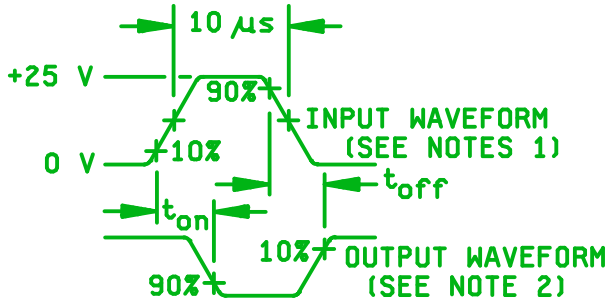
Ltr	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
CH	4.83	5.08	.190	.200
TT	0.89	1.14	.035	.045
LO	3.05 BSC		.120 BSC	
TL	16.38	16.89	.645	.665
BL	10.41	10.92	.410	.430
BLO	15.06	15.42	.593	.607
LS	2.54 BSC		.100 BSC	
TW	10.41	10.67	.410	.420
LL	26.21	28.75	1.032	1.132
MHO	13.39	13.64	.527	.537
MHD	3.56	3.81	.140	.150
LD	0.64	0.89	.025	.035
Term 1	Base			
Term 2	Collector			
Term 3	Emitter			

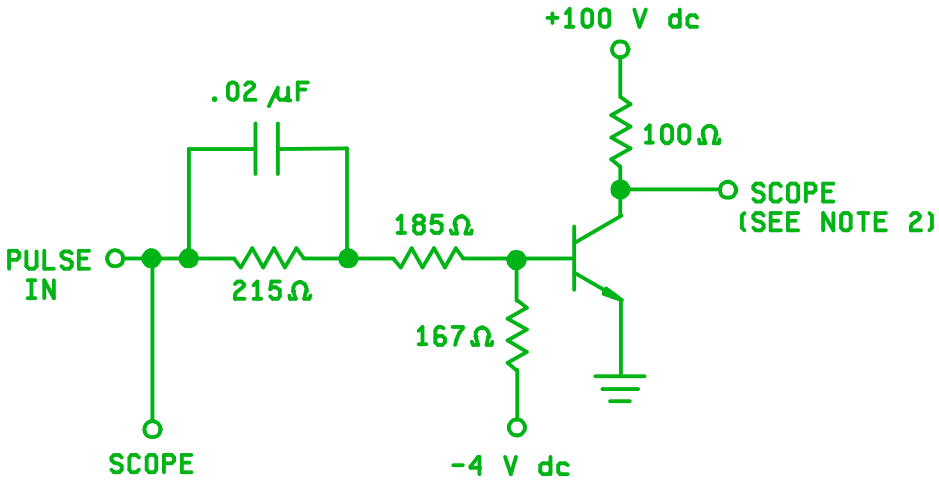


NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Glass meniscus included in dimension D and D_1.

FIGURE 2. Dimensions and configuration (T0-257AA).

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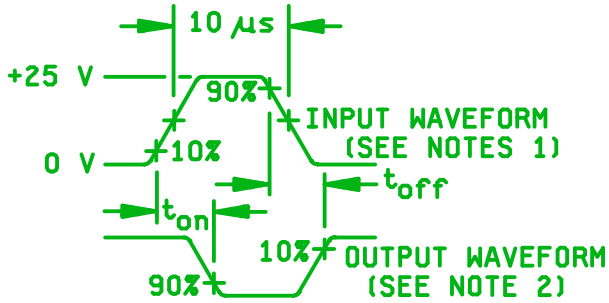


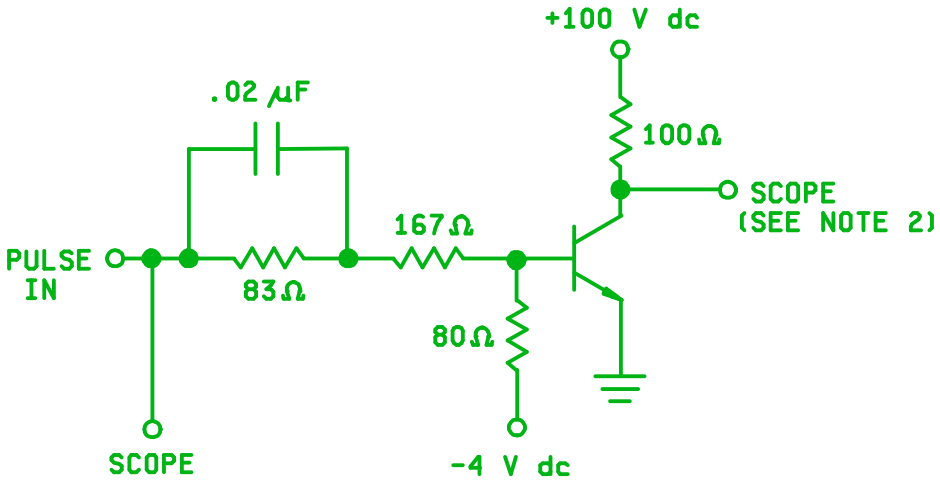
NOTES:

1. The input waveform is supplied by a pulsed generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\ \Omega$, PW = 10 µs, duty cycle ≤ 2%.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 1.0$ ns, $Z_{IN} \geq 10\ M\Omega$, $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional bypassing in order to minimize ringing.
5. The input pulse voltages and supply voltages (-4 V dc and +100 V dc) are nominal and shall be adjusted to obtain $I_{B1} = -I_{B2} = 30$ mA and $I_C = 1$ A with .02 µf capacitor removed.

FIGURE 3. Pulse response test circuit for types 2N7374 and 2N7376.

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NOTES:

1. The input waveform is supplied by a pulsed generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\ \Omega$, PW = 10 µs, duty cycle ≤ 2%.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 1.0$ ns, $Z_{IN} \geq 10\ M\Omega$, $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional bypassing in order to minimize ringing.
5. The input pulse voltages and supply voltages (-4 V dc and +100 V dc) are nominal and shall be adjusted to obtain $I_{B1} = -I_{B2} = 50$ mA and $I_C = 1$ A.

FIGURE 4. Pulse response test circuit for types 2N7375 and 2N7377.

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FIGURE 5. Maximum safe operating graph (continuous dc) for types 2N7374, 2N7375, 2N7376, and 2N7377.

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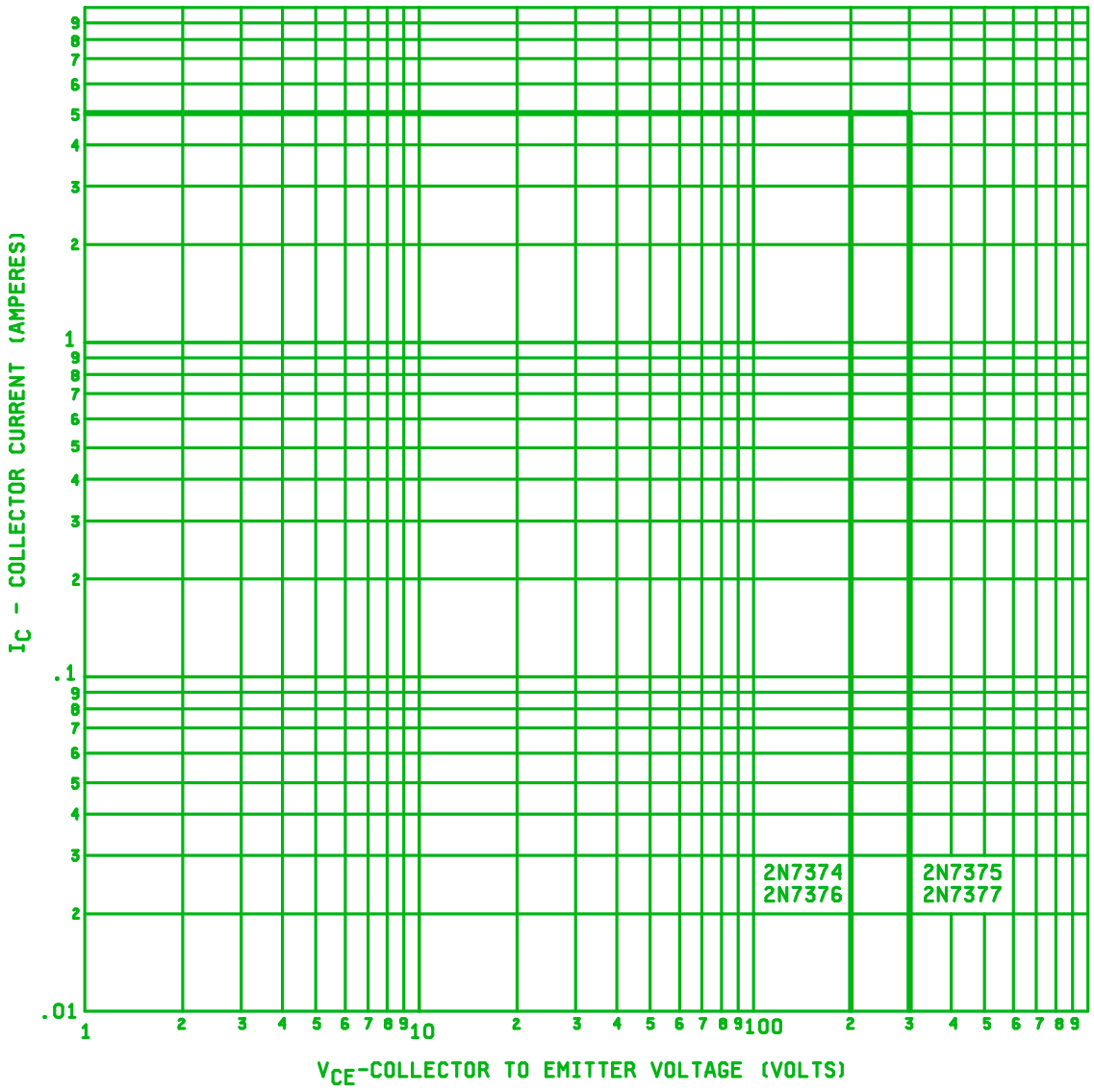


FIGURE 6. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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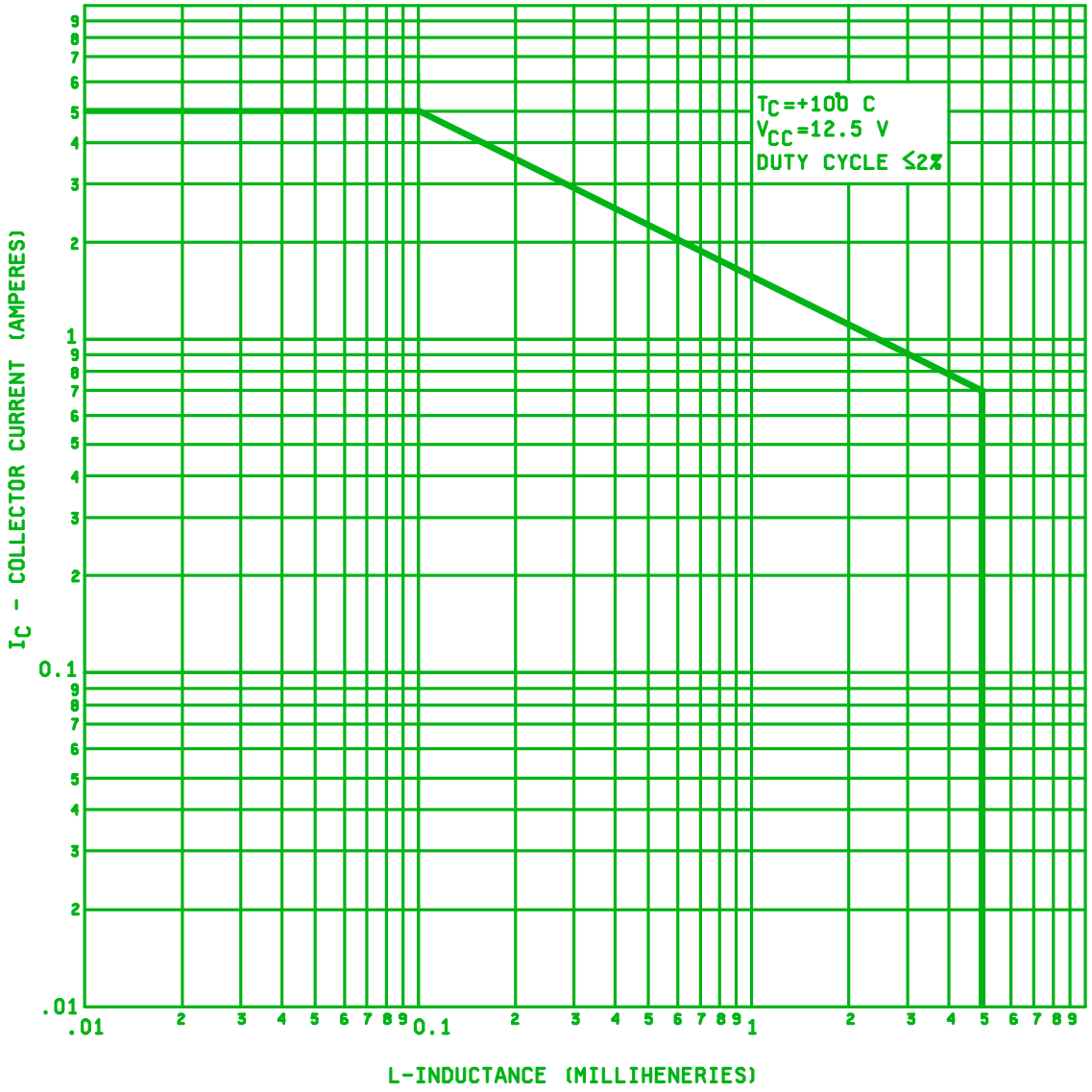


FIGURE 7. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Interchangeability information. MIL-PRF-19500/611 is a T0-254 package version of MIL-PRF-19500/455, which is a T0-213AA (T0-66) package version. The military 2N7374, 2N7375, 2N7376, and 2N7377 contains the same die as the military 2N5664 and 2N5665. The MIL-PRF-19500/611 is preferred over the MIL-PRF-19500/455 whenever interchangeability is not a problem. For new design use 2N7374 through 2N7377. The 2N5664 and 2N5665 are inactive for new design.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1902-07)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/611A	**2. DOCUMENT DATE (YYMMDD)** 970528

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER SWITCHING, TYPES 2N7374, 2N7375, 2N7376 AND 2N7377 JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (614)692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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